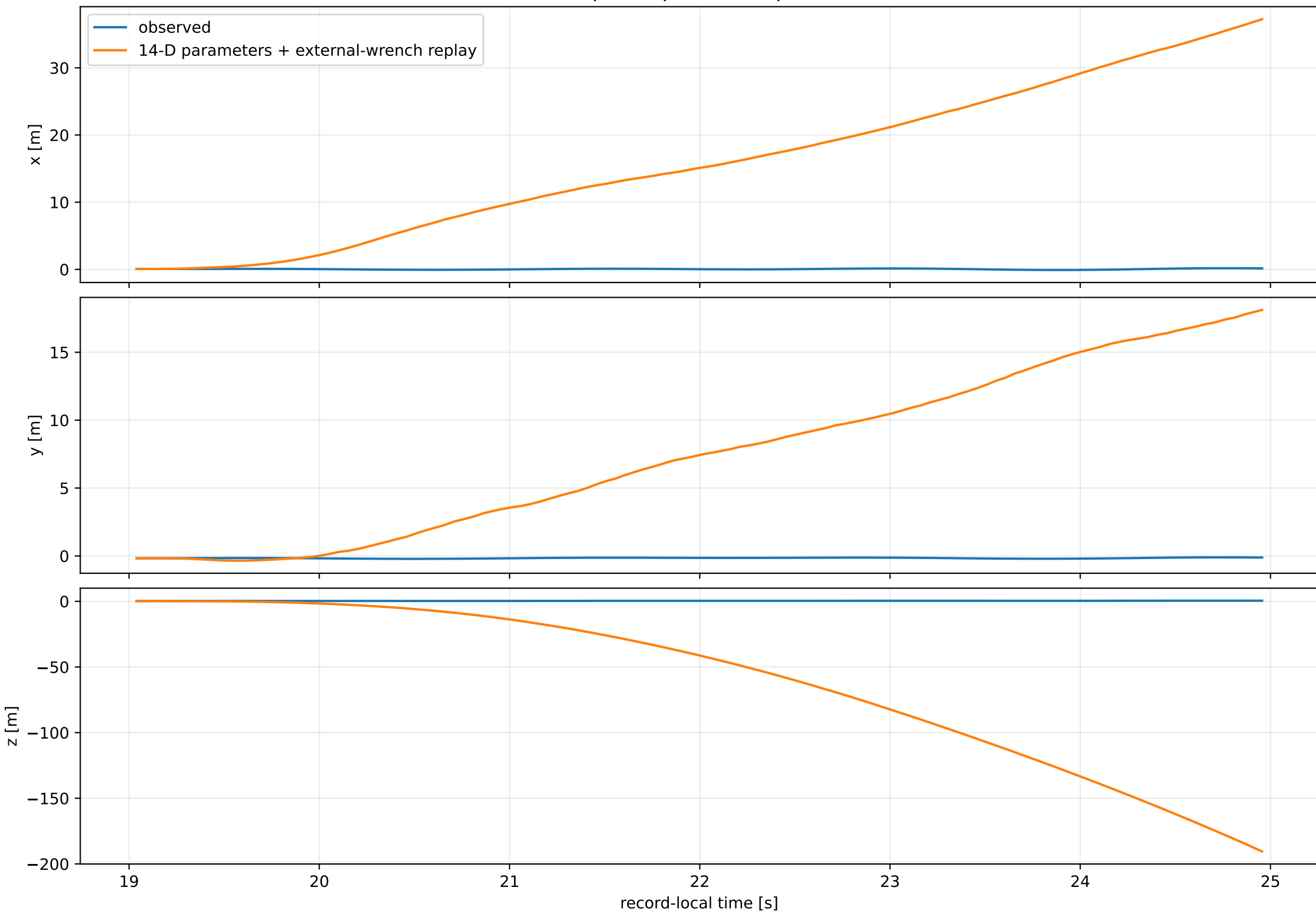
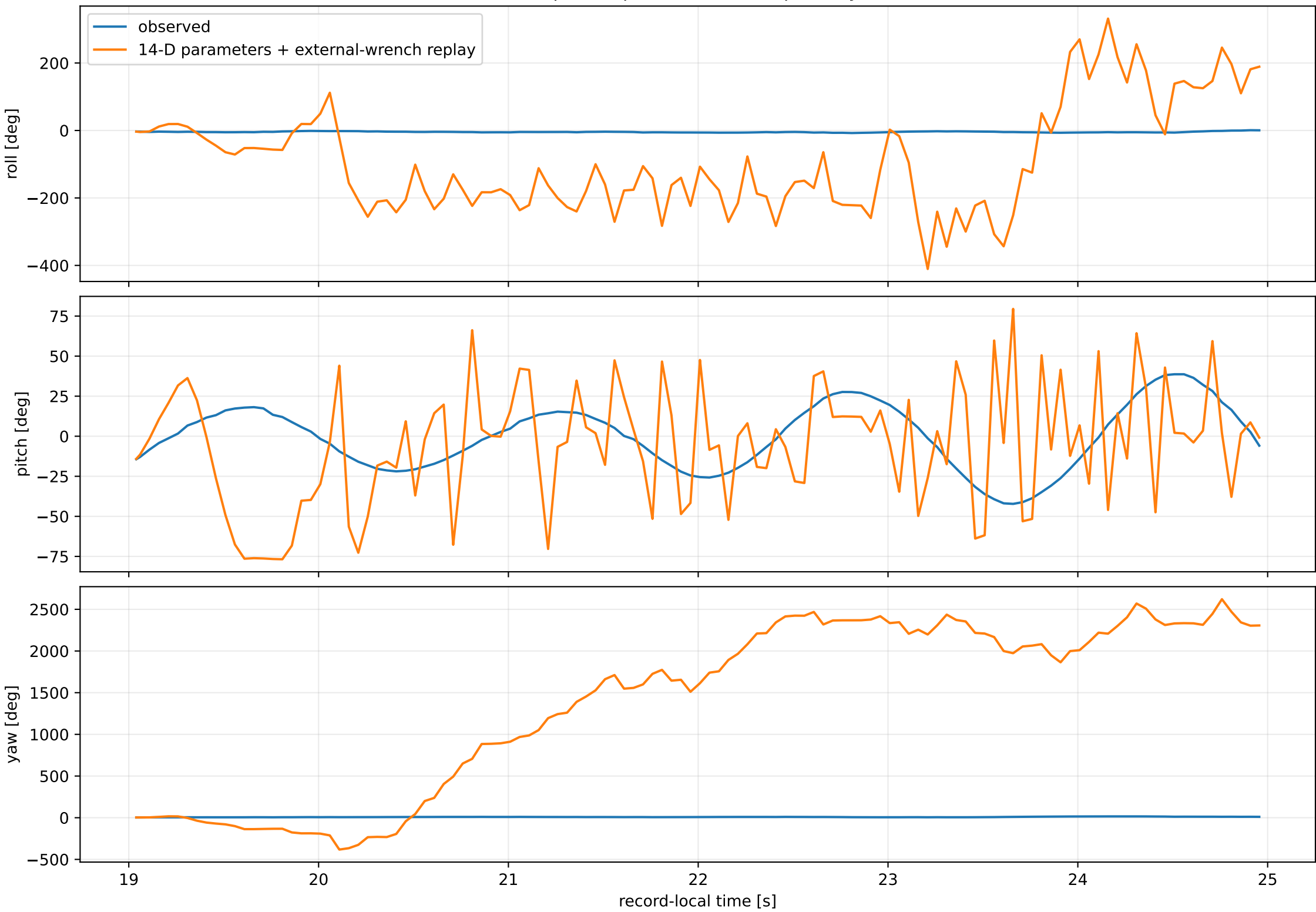


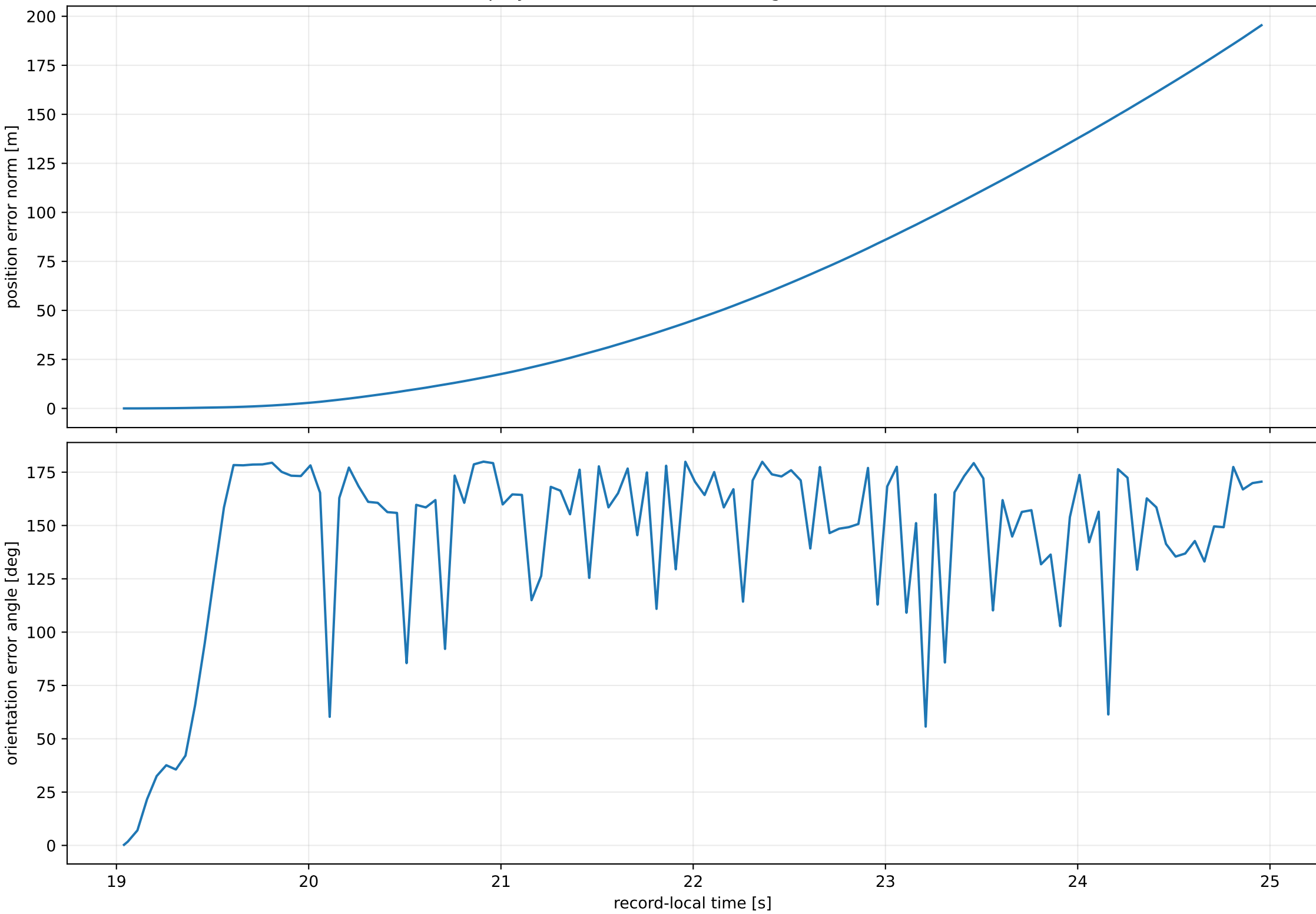
Fine pose reproduction: position



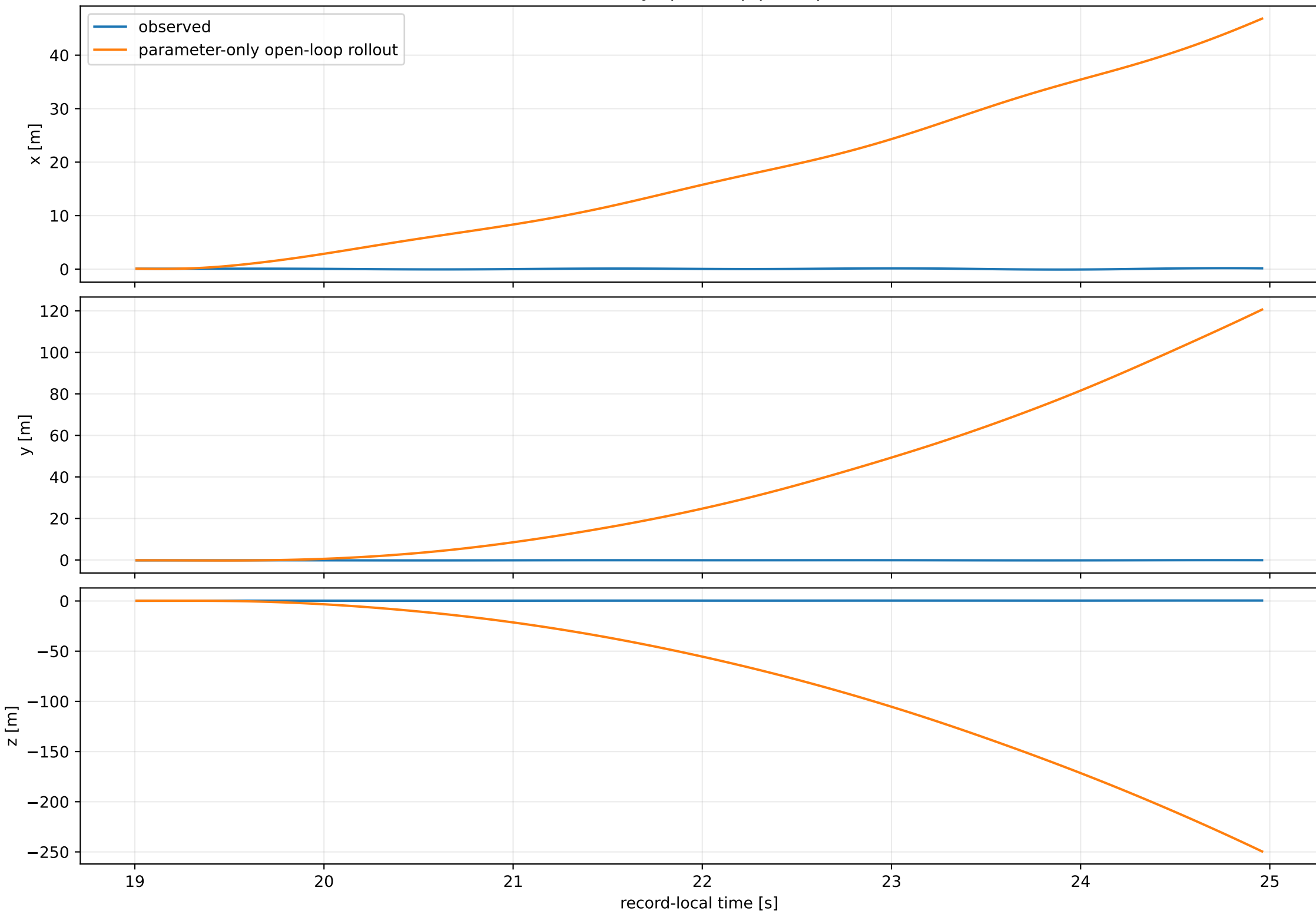
Fine pose reproduction: roll / pitch / yaw



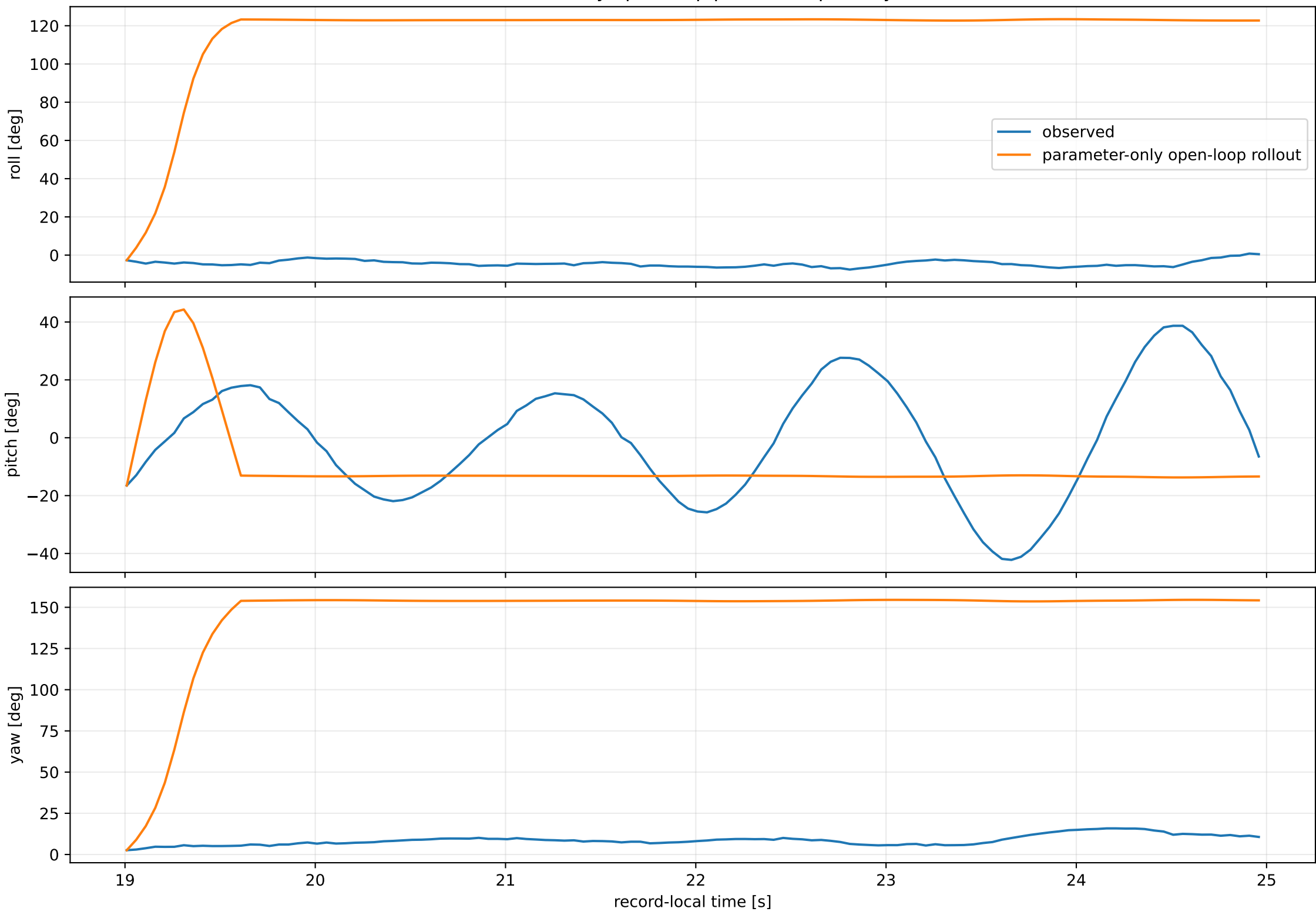
Fine replay error after the 14-D change — lower is better



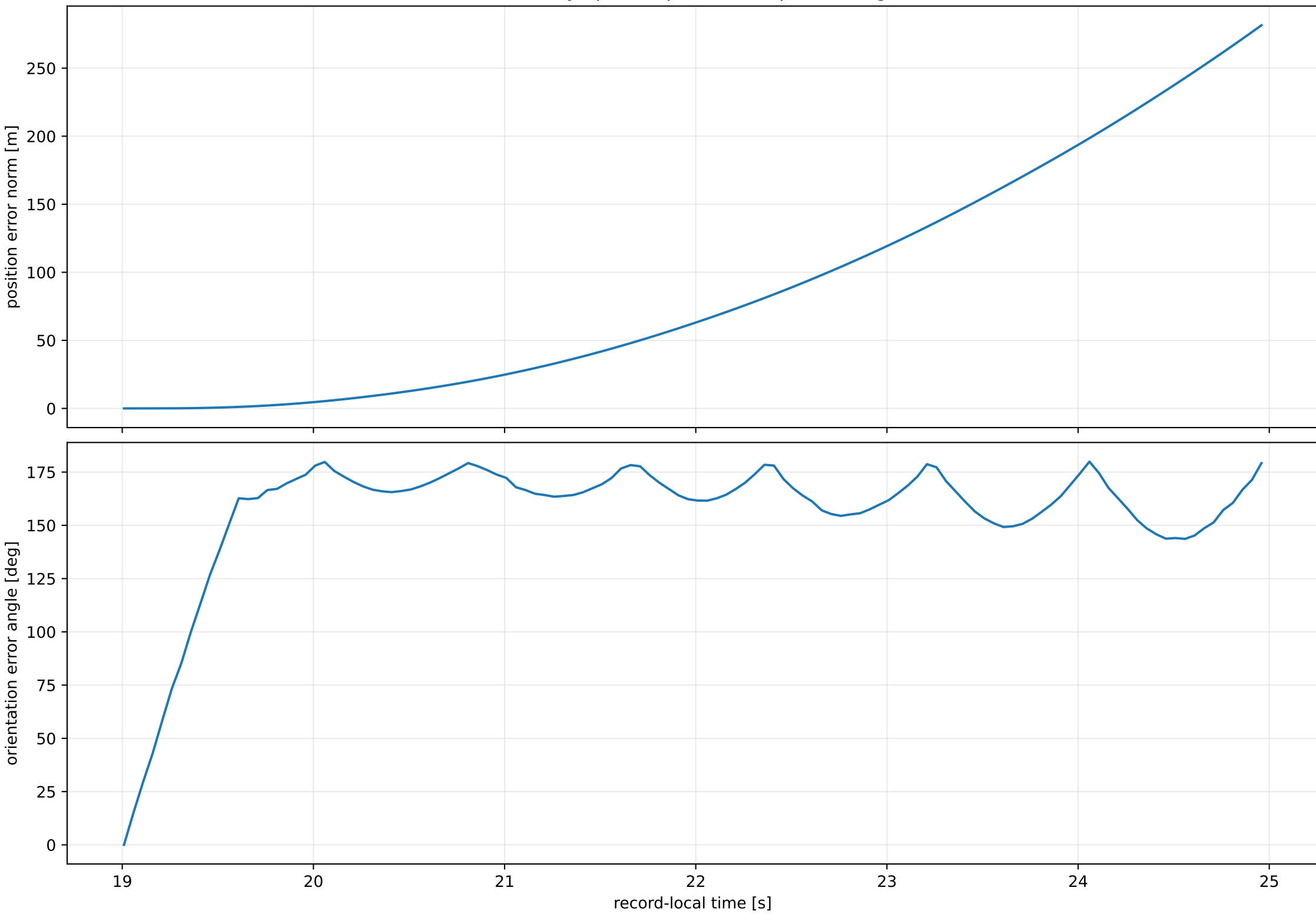
Parameter-only open-loop pose: position



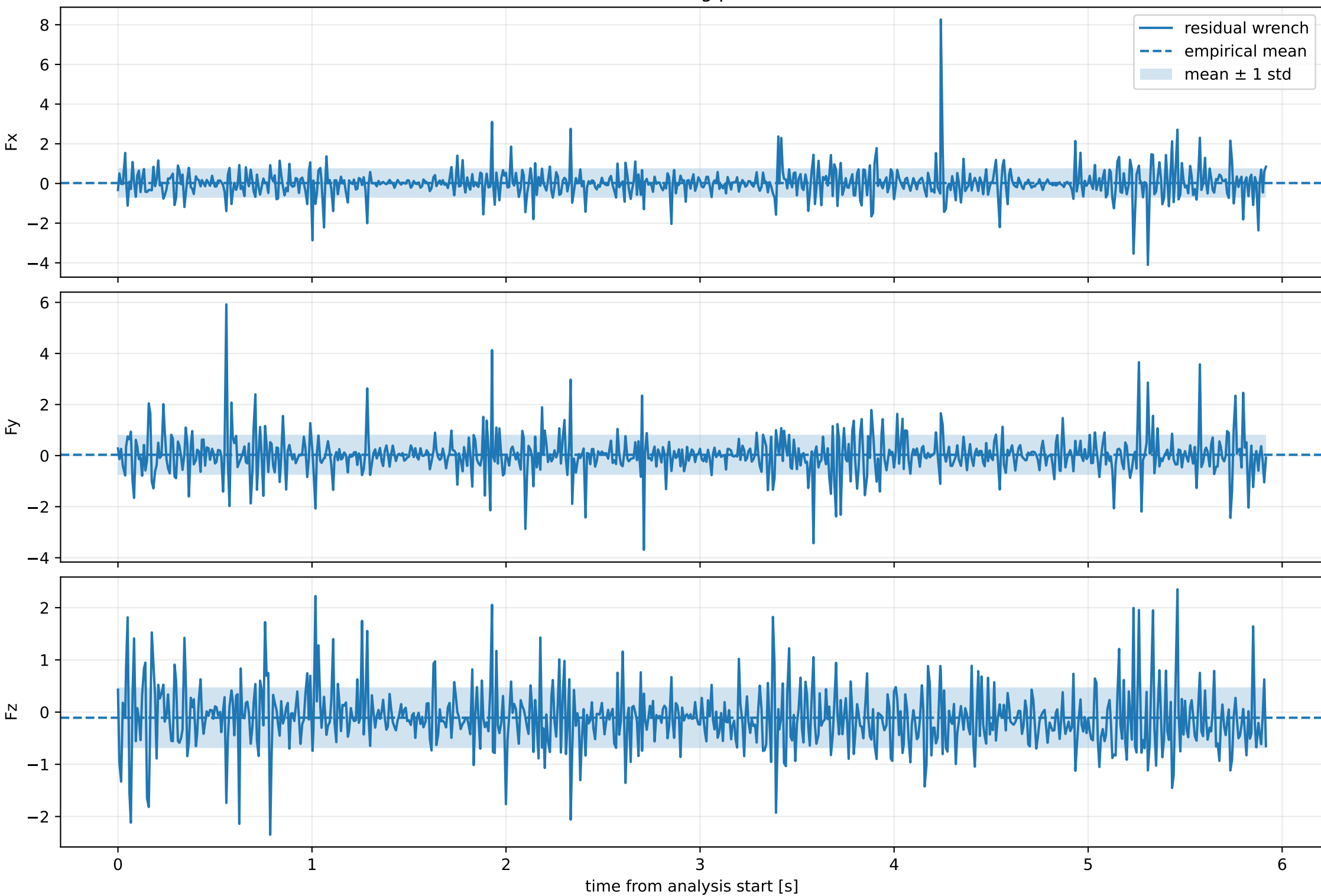
Parameter-only open-loop pose: roll / pitch / yaw



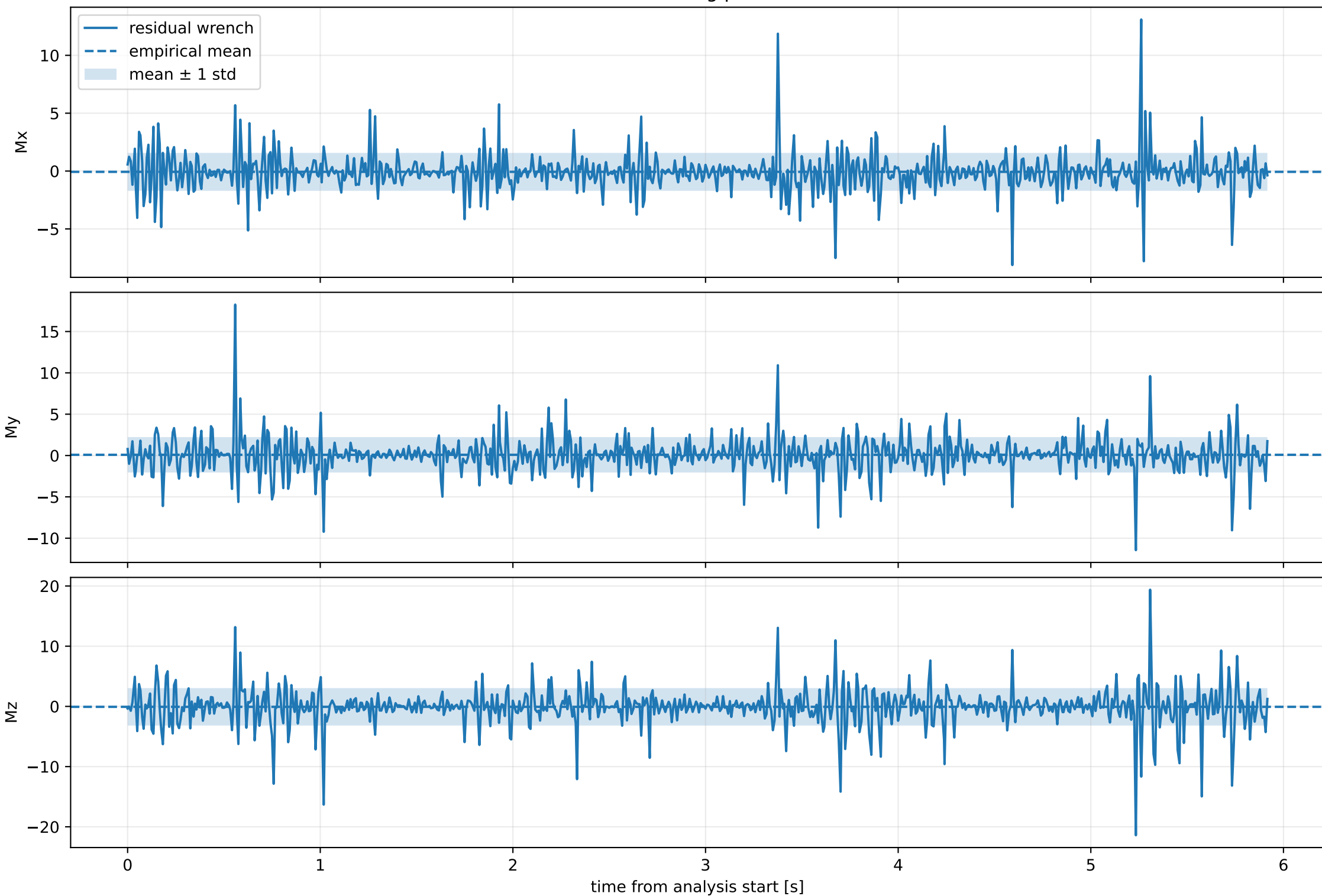
Parameter-only open-loop rollout (separate diagnostic)



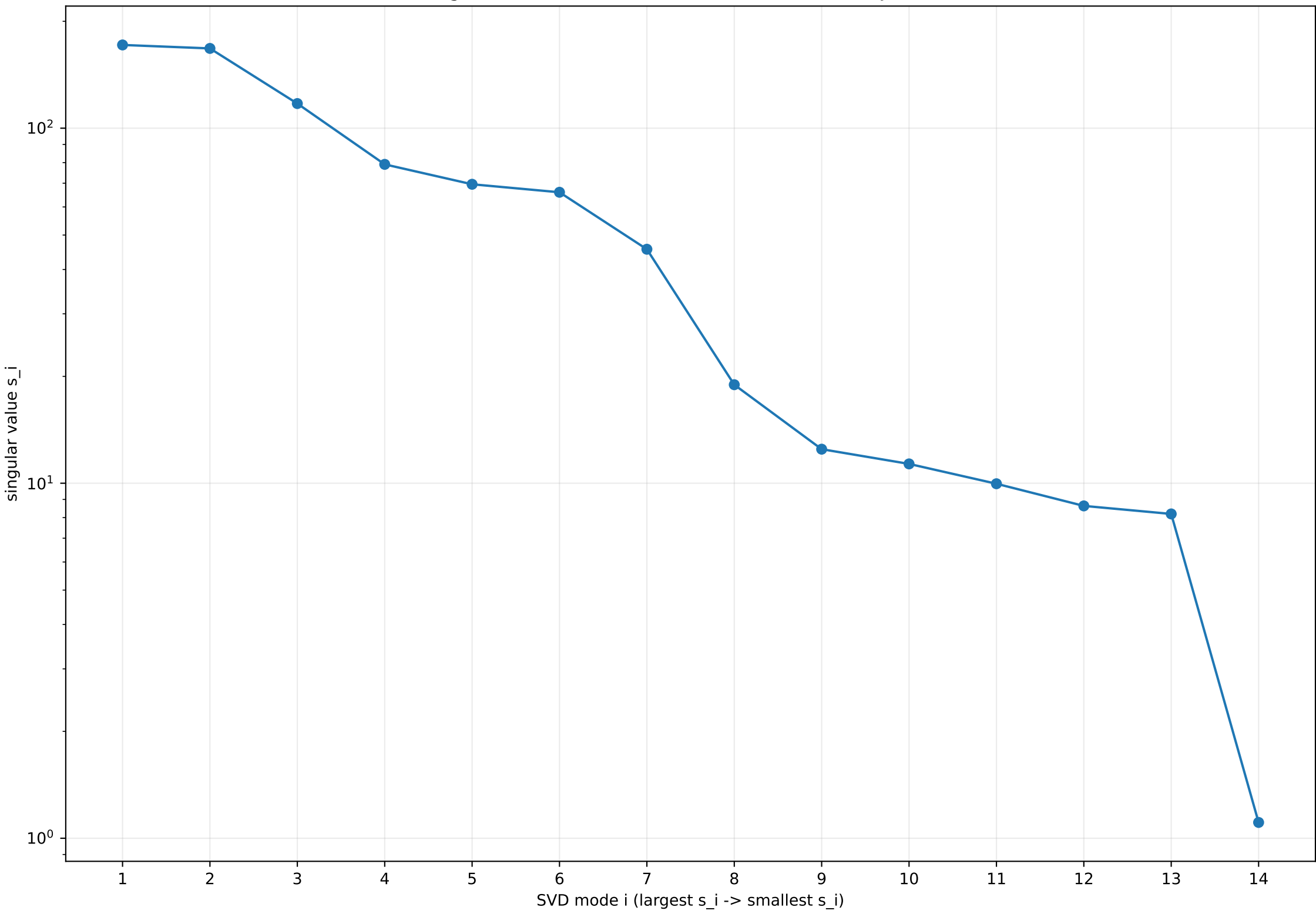
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



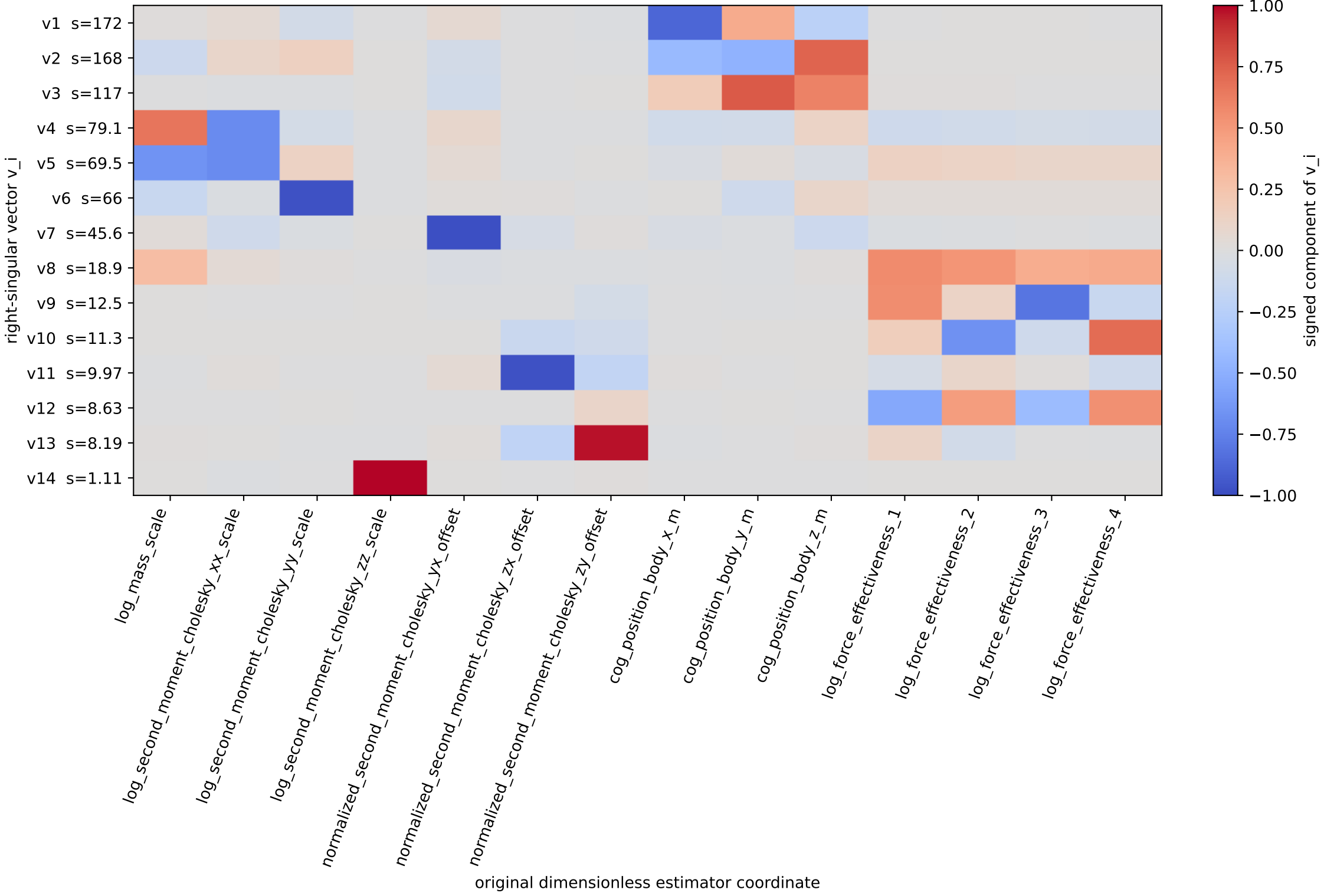
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36463251839

inertia J [kg m²]

[0.065127086 -0.00081755009 -0.00014207975]

[-0.00081755009 0.06594045 -0.0006452595]

[-0.00014207975 -0.0006452595 0.12877857]

principal moments [kg m²]

[0.064617341, 0.066443289, 0.12878547]

CoG position in body frame [m]

[-0.023521228, -0.027893487, 0.04764611]

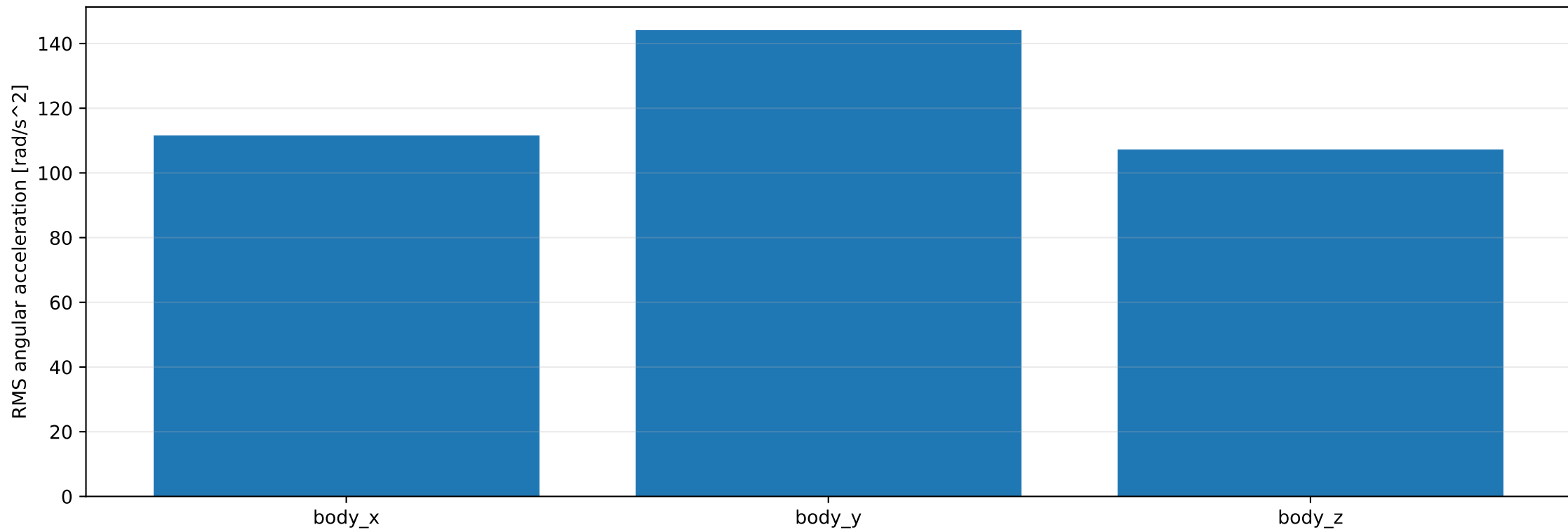
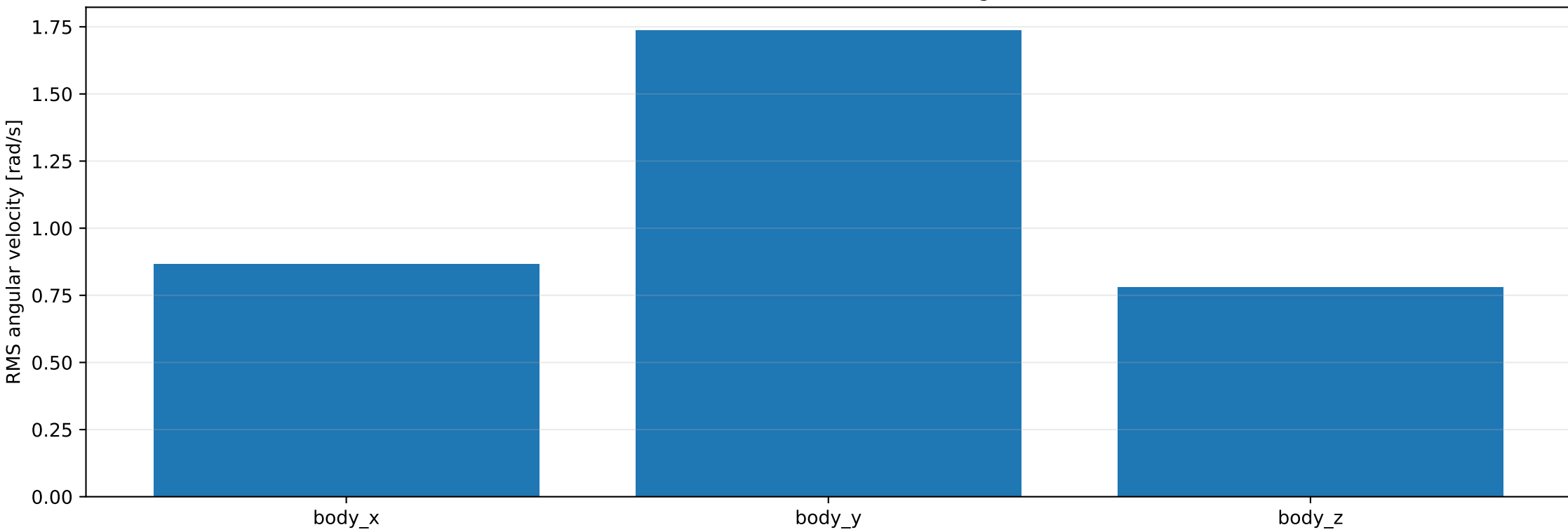
rotor force effectiveness

[1.0351462, 0.96488219, 0.88963998, 0.95120552]

selected delay [s]

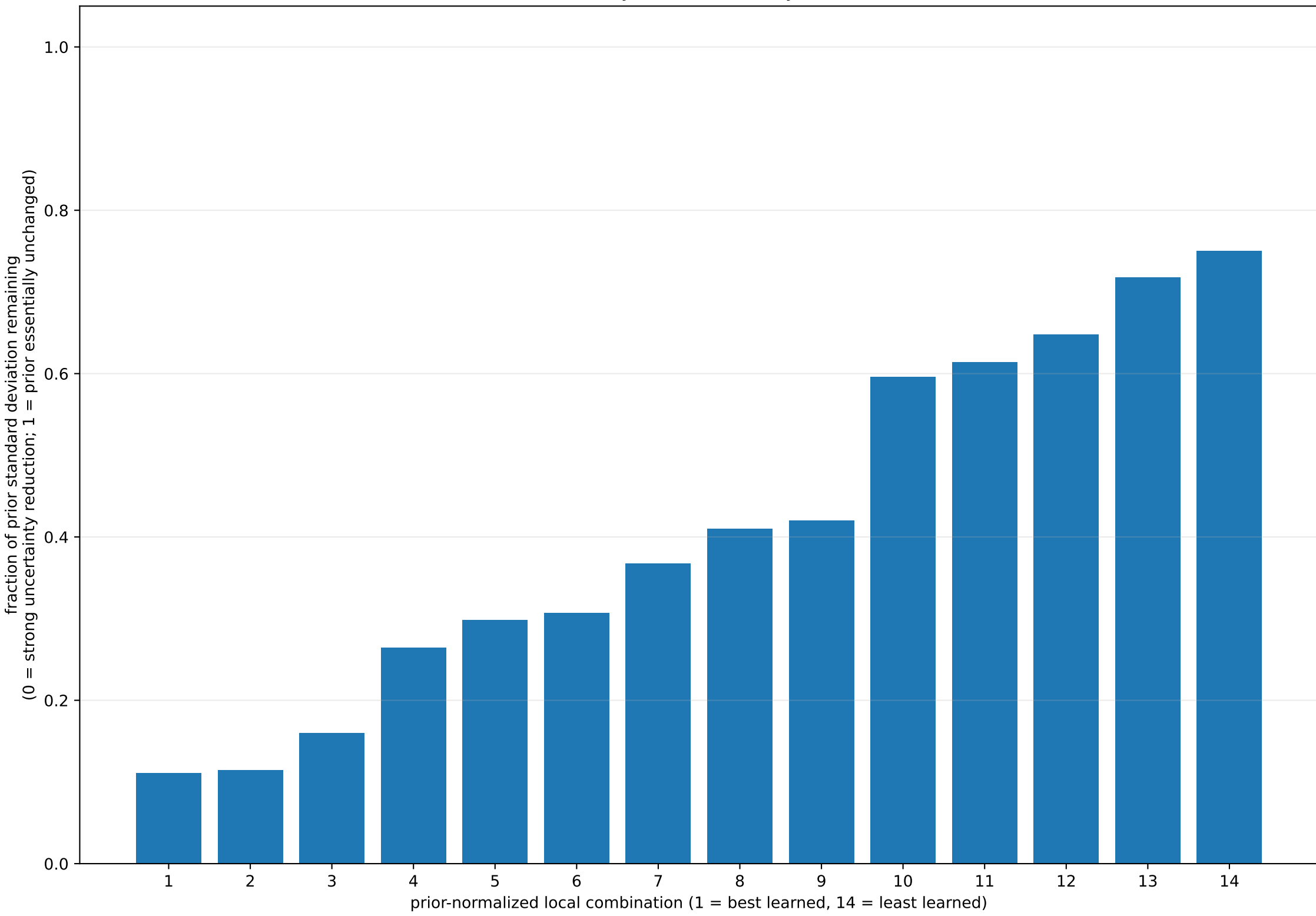
0.199136850355

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

